

Creatinine clearance (CrCl) is a measurement commonly used to estimate plasma clearance, which is the ability of the kidney to remove substances from the bloodstream for excretion. CrCl can vary based on patient health status, as shown in the table below.

Patient kidney status	CrCl (mL/min/1.73m ²)
Normal Kidney function	80-120
Depressed kidney function	<70
Elevated kidney function	>130

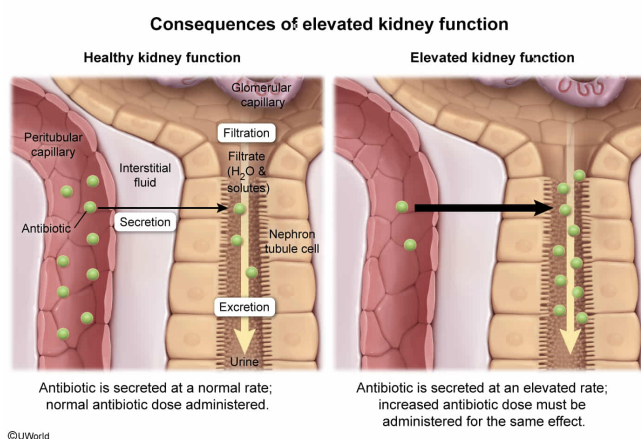
The amount of antibiotic given to a patient with elevated kidney function who has a bacterial infection, compared to the amount given to a patient with normal kidney function, should be:

- ☐ A. the same. [5%]
- ☐ B. decreased. [18%]
- ✓ ☒ C. increased. [70%]
- ☐ D. increased, but the antibiotic will have a greater effect. [5%]

Correct

70% answered correctly

Explanation:



The process of **urine formation** occurs in the **kidney**, which is divided into small functional units called **nephrons** that filter blood. This filtered blood creates a filtrate in the nephron that is ultimately concentrated into urine. Accordingly, urine formation includes production of filtrate, reabsorption of necessary solutes back into the blood, and secretion of waste products (eg, drugs such as antibiotics) into the filtrate.

Secretion of waste products occurs along the length of the nephron, but most secretion activity occurs in the proximal and distal convoluted tubules. Many waste products are secreted via [active transport mechanisms](#).

The question states that plasma clearance is the ability of the kidney to remove substances from the bloodstream. Plasma clearance is estimated by using creatinine clearance (CrCl) as a guide, with a greater CrCl indicating greater plasma clearance. The question table gives CrCl data for a patient with normal kidney function, a patient with depressed kidney function, and a patient with elevated kidney function. Compared to healthy kidney function, depressed kidney function results in reduced CrCl and elevated kidney function results in increased CrCl.

Due to *increased* CrCl in a patient with elevated kidney function, substances (eg, drugs) are cleared from the kidney at a *faster* rate than they are in a healthy (ie, normal kidney function) patient. Consequently, if this patient is given an antibiotic, the drug will likely be secreted into the nephron filtrate and cleared by the kidneys at a faster rate. Therefore, compared to a healthy patient, a patient with elevated kidney function requires an **increased** (*not* decreased) amount of antibiotic for it to have the same effect (**Choices B and D**).

(Choice A) If a patient with elevated kidney function were given the same amount of antibiotic as a healthy patient, the antibiotic would have a *decreased* effect because it would be cleared by the kidneys at a *faster* rate. Therefore, a patient with elevated kidney function requires an *increased* amount of antibiotic to achieve the same effect as in a healthy patient.

Educational objective:

During urine formation, the nephrons filter blood, reabsorb necessary solutes back into the blood, and secrete unfiltered waste products.

Time spent:0 sec

Subject:
Biology

QID:403242

System:
Digestion and Excretion